

Shayan Hussain

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PROFESSIONAL SUMMARY

Recent Computer & Information Systems Engineering graduate with a proven track record of deploying production-grade RAG pipelines, LLM integrations, and MLOps workflows. Seeking an Applied AI role (onsite, remote, or hybrid) to drive scalable, real-world automation through agentic systems and full-stack development.

EDUCATION

NED University of Engineering and Technology	Karachi, Pakistan
Bachelor of Engineering in Computer and Information Systems Engineering	2022 – 2026
CGPA: 3.556 / 4.0 Latest SGPA: 3.95 / 4.0	

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, Dart, HTML, CSS, SQL

AI & Machine Learning: CRAG (Corrective RAG), Ragas, LlamaIndex, RAG Pipelines, LLMs, LangChain, LangGraph, Vector Search, MLOps, TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLO, CNNs, NLP, Transformers, Reinforcement Learning, Pandas, NumPy

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Containerization, Nginx, GitHub Actions (CI/CD), GHCR, Vercel, Render

Frameworks & Tools: FastAPI, Fastify, Next.js (App Router), React, React Flow, Node.js, BullMQ, Celery, Drizzle ORM, TailwindCSS, shadcn/ui, Django, Flask, MERN Stack, Flutter, MCP, FastMCP, n8n, Make.com, UptimeRobot, Git, Vitest, Langfuse

Databases & Storage: PostgreSQL, pgvector, Redis (Semantic Caching), MySQL, Supabase, SQLite

Core Competencies: Evals-Driven Development, Agentic AI, Multi-Tenancy Architecture, Production Deployment, Computer Vision, Full-Stack AI SaaS, Exactly-Once Processing, Distributed Task Queues

WORK EXPERIENCE

Shispare	Karachi, Pakistan
<i>AI Engineer (Full-time)</i>	January 2026 – May 2026

- Designed and deployed a production-grade RAG system for the Stellar LMS platform, successfully automating the generation of assessment questions from PDF course content in a live environment.
- Engineered an operational Model Context Protocol (MCP) server for Pipedrive CRM within a 4-person team, automating workflows to eliminate manual effort by 50% and achieving single-digit second response times for AI agent integrations.
- Architected an agentic coding assistant via a VS Code extension, utilizing locally deployed LLMs and a centralized RAG index over Bitbucket/GitHub to ensure enterprise data privacy while accelerating developer workflows.
- Optimized and deployed local LLMs, reducing inference and response times by up to 30% to decrease API dependency and enhance internal tooling efficiency.

10Pearls	Remote
<i>MERN Stack Intern</i>	September 2025 – November 2025

- Built a scalable, production-ready Notes Application utilizing the MERN stack with robust CRUD functionality, achieving 90%+ test coverage through rigorous Jest unit testing.
- Partnered directly with a senior engineer to execute asynchronous development cycles, managing modular feature branches via GitHub to support seamless remote delivery.

- Contributed to the development of a React and FastAPI-based Art DRM Marketplace, successfully integrating PayPal to process secure payment gateways.
- Prototyped AI-based virtual Try-On technology integration approaches, validating technical feasibility for commercial product fitting use cases.

TECHNICAL PROJECTS

Ringback — Text-First AI Voice Agent Platform

2026

Python, FastAPI, Next.js 14, PostgreSQL, Supabase, Make.com, Docker | [Live Demo](#) | [GitHub](#)

- Architected a channel-agnostic AI voice agent core via FastAPI, achieving 100% task success and a 0% false-action rate across 14 CI-enforced evaluation scenarios.
- Designed deterministic rule-based NLU for reproducible regression testing, with hot-swappable Groq/Gemini integration for production deployment.
- Integrated Make.com with HMAC-signed webhooks to automate event routing (bookings, escalations, leads) to Google Calendar and Gmail with sub-second latency.
- Deployed a multi-tenant production stack using Vercel, Render, and Supabase, implementing JWT auth, strict tenant-scoped data isolation, and encrypted credential storage.

Clause — Evals-First Vertical Document Intelligence System

2026

Next.js 16, FastAPI, LangGraph, LlamaIndex, pgvector, Supabase, Ragas | [Live Demo](#) | [GitHub](#)

- Architected a Corrective RAG (CRAG) legal contract engine using LangGraph and FastAPI, incorporating multi-hop reasoning and a self-critic node to achieve >95% faithfulness and eliminate untraceable hallucinations.
- Pioneered an evals-first methodology via Ragas, gating architectural changes behind an ablation table to drive a 40% increase in multi-hop accuracy compared to base RAG baselines.
- Engineered a structure-aware ingestion pipeline to *pgvector*, linking embeddings to exact hierarchical clause IDs (e.g., 8.1a) to guarantee 100% citation precision and data provenance.
- Developed a Next.js 16 split-view UI featuring verified citation chips that auto-scroll to the exact source text, ensuring dynamic abstention policies when retrieval fails.

Consensus — Supervisor-Orchestrated Multi-Agent Platform

2026

Next.js 14, FastAPI, Python, LangGraph, Redis, PostgreSQL, FastMCP, Docker | [Live Demo](#) | [GitHub](#)

- Developed a stateful, supervisor-orchestrated multi-agent system using LangGraph and FastAPI, managing specialized agents (Researcher, Analyst, Writer) and enforcing a budget ceiling of \$1.50 per run and a 15-step loop ceiling.
- Engineered a robust Human-in-the-Loop (HITL) gate that halts graph execution for consequential actions, serializing state to PostgreSQL with sub-50ms latency and resuming via Redis job queues.
- Integrated Langfuse for execution tracing, and built an independent, append-only PostgreSQL audit log guaranteeing 100% traceability of all agent tool invocations and user approvals.
- Created a Next.js console displaying real-time agent run status, workspace-level plan limits (50 runs/mo), and a workflow approval dashboard.

Flowlet — AI-Native Workflow Automation Platform

2026

Next.js 14, Fastify, PostgreSQL, BullMQ, Redis, React Flow, Docker | [Live Demo](#) | [GitHub](#)

- Architected a distributed, multi-tenant AI workflow automation engine utilizing Next.js, Fastify, and BullMQ, enabling users to visually compose complex *trigger* → *AI-step* → *branch* pipelines.
- Engineered a highly concurrent job execution worker pool featuring 3-layer idempotency and crash-safe fairness leases, guaranteeing exactly-once execution and maintaining a flat ~20ms p95 enqueue latency under 500+ concurrent burst loads.

- Implemented robust backend hardening, including Nginx edge rate-limiting (50 req/s), per-workspace LLM cost tracking, and an AI output cache that eliminated 100% of redundant API token costs for duplicate inputs.
- Validated system reliability through an automated load-testing artifact, achieving 96 req/s sustained throughput and seamlessly draining 500 concurrent triggers in under 14 seconds with zero dropped runs or double-executions.

Deflekt — Multi-Tenant AI Support Deflection SaaS

2026

AWS EC2/RDS, Next.js 14, FastAPI, pgvector, Redis, Celery, Docker | [Live Demo](#) | [GitHub](#)

- Architected a multi-tenant RAG platform using Next.js 14, Python FastAPI, PostgreSQL (pgvector), and Redis, reducing repetitive support ticket volume by 70% with cited, grounded answers.
- Engineered a sub-18ms Redis semantic cache cutting duplicate LLM query latency by 98.5% and scaled single-instance AWS EC2 throughput to 150+ req/sec (cached) and 45+ req/sec (streaming RAG), supporting 10,000+ daily active users.
- Deployed an async Celery worker queue processing 1,200 words/sec for document ingestion and built an automated CI/CD pipeline on AWS EC2/RDS with an LLM-as-a-Judge eval harness gating builds on >80% Faithfulness precision.
- Enforced strict database-level multi-tenancy isolation and an automated confidence gate (<0.72) that escalates low-confidence queries to human support with <0.8% hallucination rate.

RoboComm: Smart Interactive Campus Robot

2025 – Present

- Built an autonomous campus assistant using a custom ESP32 chassis integrated with an MPU6050 (heading estimation) and HC-SR04 (obstacle detection), achieving a navigation accuracy of 85%.
- Deployed real-time YOLO object detection maintaining 25–30 FPS and integrated the Gemini API for natural language STT/TTS interaction, enabling voice-driven campus navigation.

Voice-Enabled Booking Automation System

2026

- Architected a zero-touch conversational agent utilizing Make.com workflows to orchestrate Calendly, Google Calendar, and OpenAI APIs, streamlining the booking lifecycle to a sub-5-minute process and successfully processing 100+ automated user interactions.

Anomaly Detector Computer Vision Pipeline

2026

- Developed an end-to-end MLOps pipeline (hand-annotating 1,500 images via CVAT and augmenting via Roboflow) to train an RT-DETR-L model with a frozen backbone, achieving 88% accuracy and a peak mAP of 0.650 for anomaly detection on unseen datasets.

Remote Work Productivity Tracker

2026

- Trained a behavioral classification model on a custom, annotated screen-recording dataset of 1,000 hours to track verified work hours with 80% accuracy, building a real-time Next.js/Node.js analytics dashboard for administrative oversight.